INVENTORY MODELS

Inventory modelling deals with determining the level of a commodity that a business must maintain to ensure smooth organization. Inventories are vital to the successful functionary of manufacturing and retailing organizations. The inventories may consist of raw materials, work-in-progress, spare parts/ consumers, and finished goods.

Inventories may be held for a variety of purpose, but in general, there are five types of inventories that an organization can use for serving these purposes. These are:

a) Movement inventories
   Also called transit or pipeline Inventories. Their existence owes to the fact that transportation time is involved in transferring substantial amounts of resources.
b) Buffer inventories
   Are held to protect against the uncertainties of demand and supply. An organization generally knows the average demand for various items that it needs. However, the actual demand may not exactly match the average and could well exceed it. Safety stocks are held for protecting against the fluctuations in demand and lead time. The idea of keeping buffer stocks is to render a higher level of customer service and consequently reduce the number of stockouts and back-orders.
c) Anticipatory Inventories
   They are held for the reason that a future demand for the product is anticipated. The idea underlying it is to smoothing the production process producing for a longer duration on a continuous scale rather than operating with excessive overtime on one period and then let the system be idle or close down for the reason of inadequate demand for another period.
d) Decoupling Inventories
   The idea is to decouple or disengage different parts of the production system. Clearly the decoupling inventories act a shock absorbers and have cushioning effect in the face of varying work-rates, and machines breakdowns and failures and so on.

e) Cycle inventories
   They are held for the reason that purchase are usually made in lots rather than the exact amounts which may be needed at a point in time. As the cycle inventories are related to the purchases in lots, they are also called lot-size inventories and the larger the lot size the greater would be the level of cycle inventories.

General Inventory (Model) Decision

In an inventory control situation, there are three basic questions to be answered. They are:

1) How much to order?
2) When to order?
3) How much safety stocks should be kept?

The basis for answering these questions is the minimization of the following inventory cost functions:

$$(Total\ Inventory\ cost) = (Purchasing\ cost) + (Setup\ cost) + (Holding\ cost) + (Shortage\ cost)$$

Inventory Management Systems

There are two inventory management systems. They are:

a) Fixed Order Quantity System (or Reorder Point System)
In which a level called the reorder level is determined. As soon as the stock level reaches this point, an order for a predetermined number of units is placed.

b) Periodic Review System
Where the inventories are replenished at fixed intervals of time. Whereas the size of order is fixed in the previous system and the time is not, in this system the time after which the supplies are ordered is fixed but not the quantity to be ordered.

Demand and Inventory Models

The demand patterns in an inventory model may assume one of four types:

1) Deterministic and constant (static) with time
2) Deterministic and variable (dynamic) with time
3) Probabilistic and stationary over time
4) Probabilistic and nonstationary over time

To determine if a certain approximation of demand is acceptable, we compute the coefficient of variation (CV) and then use it to determine the nature of demand using the following guidelines:

1) If the average monthly demand is 'approximately' constant for all months and CV is reasonably small (<20%). Then the demand may be considered deterministic and constant with its value equal to the average of all monthly demands
2) If the average monthly demand varies appreciably among the different months but CV remains reasonably small, then the demand is considered deterministic but variable.
3) If, in case 1, CV is high (>20%) but approximately constant, then the demand is probabilistic and stationary
4) Otherwise, it is probabilistic and non-stationary.

Example: Analyse the nature of the following demand for product X:

	WEEKLY DEMAND			
MONTHS	1	2	3	4
JAN	43	44	45	68
FEB	69	90	41	79
MAR	50	76	34	81
APR	67	80	39	22
MAY	27	86	41	80
JUN	34	74	40	20

Solution

		WEEKLY DEMAND						
	MONTHS	1	2	3	4	mean	Standard deviation	Coefficient of Variation
	JAN	43	44	45	68	**50.00**	**12.03**	**24.06**
	FEB	69	90	41	79	**69.75**	**21.00**	**30.10**
	MAR	50	76	34	81	**60.25**	**22.16**	**36.77**
	APR	67	80	39	22	**52.00**	**26.32**	**50.61**
	MAY	27	86	41	80	**58.50**	**28.97**	**49.51**
	JUN	34	74	40	20	**42.00**	**22.92**	**54.57**
mean		**48.33**	**75.00**	**40.00**	**58.33**			
Standard deviation		**17.13**	**16.33**	**3.58**	**29.30**			
Coefficient of Variation		**35.44**	**21.78**	**8.94**	**50.23**			

SUMMARY	*Count*	*Sum*	*Average*	*Variance*
JAN	4	200	50	144.6667
FEB	4	279	69.75	440.9167
MAR	4	241	60.25	490.9167
APR	4	208	52	692.6667
MAY	4	234	58.5	839
JUN	4	168	42	525.3333
wk1	6	290	48.33333	293.4667
wk2	6	450	75	266.8
wk3	6	240	40	12.8
wk4	6	350	58.33333	858.6667

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Months	1837.333	5	367.4667	1.035831	0.432306	2.901295
Weeks	4079.167	3	1359.722	3.832843	0.032085	3.287382
Error	5321.333	15	354.7556			
Total	11237.83	23				

Inventory Costs

In determining an optimal inventory policy, all or some of these costs components are included in the objective function:

i) Purchase Cost
This refers to the nominal cost of inventory. It is the purchase price for the items that are bought from outside source and the production cost if items are produced within the organization.

ii) Ordering Cost/Set-Up Cost
This is incurred whenever the inventory is replenished. It includes costs associated with the processing and chasing of the purchase order, transportation, inspection for quality, expediting overdue order, etc, it is also known as procurement cost.

iii) Carrying Cost
Also known as holding or storage cost, carrying cost represents the cost that is associated with storing an item in inventory. Its elements include the opportunity cost of capital of invested in the stock; the costs directly associated with storing goods, the obsolescence cost; deterioration costs and cost incurred in preventing deterioration, and fire and general insurance, etc.

iv) Stockout Cost
The cost associated with not serving the customers. Stockout imply shortages. This could be internal or external.

Fixed – Order Quality System

The classical EOQ Model (Wilson Formulation)

EOQ – Economic Order Quantity Model

Under the EOQ Model, a cost model is first developed, and then it is manipulated to form an inventory model.

Assumptions:

i. The demand for the item is certain, continuous and constant with time.
ii. The lead time, that is, the time between placing order and its delivery is known and fixed. Thus when the lead time is zero, the delivery is instantaneous.
iii. Within the range of the quantities to be ordered, the per unit holding cost and the ordering cost are constant and independent of the quantity ordered.
iv. The purchase price of the item is constant, no discount is available on purchase of large lots.
v. The inventory is replenished immediately as the stock level reaches zero. Consequently, there are no stock overages or shortages.

The cost model is

$$T(Q) = O(Q) + H(Q)$$

Where

Q – the ordering quantity

T(Q) – total (variable) annual inventory cost

O(Q) – total annual ordering cost

H(Q) – total annual holding cost

Example

Ade ltd produces 2000 TV sets in a year for which it needs an equal number of tubes of a certain type. Each tube costs N10 ('000) and the cost to hold a tube in stock for a year is N2.40.besides, the cost of placing an order is N150, which is not related to its size.

Now, if an order for 2,000 tubes is placed, only one order per annum is required. When 1000 units are ordered for, 2 orders in a year are needed, while if 500 units are ordered to be supplied, then a total of 4 orders per annum are required. Naturally, as the number of orders placed increases the ordering cost goes up. More orders, however, would also imply smaller order quantity and, therefore, decreasing holding costs. Thus we have a trade off between the ordering and the holding cost.

Find the order size that minimizes the cost function T(Q).

Solution

a) Total annual ordering cost
   This is given by the number of times an order is placed multiplied by the ordering cost per order, A.
   i.e. $O(Q) = N \times A$
   where $N = \frac{D}{Q} = \frac{annual\ demand}{order\ quantity}$
   $O(Q) = \frac{D}{Q} \times A$
   For the example, when
   N = 1, Q = 2000 and O(Q) = 1 × 150 = 150
   N = 2, Q = 1000 and O(Q) = 2 × 150 = 300
   N = 4, Q = 500 and O(Q) = 4 × 150 = 600
   N = 5, Q = 400 and O(Q) = 5 × 150 = 750
b) Total annual holding cost
   Obtained by multiplying the unit holding cost, h, by the average number of units held in the inventory, Q/2
   $H(Q) = \frac{Q}{2} \times h$
   For our example, when
   Q = 2000, $H(Q) = \frac{2000}{2} \times 2.40 = 2400$
   Q = 1000, $H(Q) = \frac{1000}{2} \times 2.40 = 1200$
   Q = 500, $H(Q) = \frac{500}{2} \times 2.40 = 600$

Q = 400, $H(Q) = \frac{400}{2} \times 2.40 = 480$

c) Total annual inventory cost (variable)
Obtained by adding up both components of (a) and (b)

Calculation of Total Cost			
Order Quantity Q	Annual Ordering Cost O(Q)	Annual Holding Cost H(Q)	Total Annual Cost T(Q) = O(Q) + H(Q)
100	3000	120	3120
200	1500	240	1740
400	750	480	1230
500	600	600	1200
1000	300	1200	1500
2000	150	2400	2550

<u>Determination of the re-order level</u>

Suppose the following additional data are given:

No. of working days in the year = 250

Lead time = 15 working days

With this information, the daily demand = 2000/250 = 8

Demand during lead time = 15 × 8 = 120

Therefore, re-order level = 120.